Mesoscopic cycles of comparable-rate Luce permutations: a switching proof of Conjecture 13.1

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Abstract

Let $X_{n,i}$ be independent exponential clocks with rates $1 \le \theta_{n,i} \le \kappa$, and let $\sigma_n(i)$ be the rank of $X_{n,i}$ among the n clocks. We prove that

$$\mathbb{E}C_{n,r_n} \sim \frac{1}{r_n} \qquad (r_n \to \infty,\ r_n = o(n)),$$

and that, for every $a_n \to \infty$ with $a_n = o(n)$, the point process of cycle lengths divided by a_n converges vaguely on $(0,\infty)$ to the Poisson point process with intensity $\mathrm{d}x/x$. The proof uses no quenched local law, no uniform regularity of the terminal clock tail, no residual joint-cycle theorem, and no macroscopic cycle-limit input. Its main ingredients are a marked involutive clock-swap identity, uniform integrability of the swap factors, and a finite-endpoint mixing theorem in mean conditional total variation relative to a coarse clock environment. The endpoint theorem is formulated with a free admissible terminal-window depth. The square-root choice suffices for qualitative mesoscopic convergence, while a logarithmic-depth specialization gives polynomial endpoint-mixing error on the logarithmic windows used in quantitative order problems. A tilted version of the one-cycle identity gives the limiting Laplace functional.

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1 Model, prior input, and main result

For each n, let

$$X_{n,i} \sim \mathrm{Exp}(\theta_{n,i}), \qquad 1 \le i \le n,$$

be independent, where throughout

$$1 \le \theta_{n,i} \le \kappa \tag{1}$$

for one fixed $\kappa < \infty$. Since the clocks are continuous, they are almost surely distinct. The associated Luce permutation is

$$\sigma_n(i) = 1 + \#\{j : X_{n,j} < X_{n,i}\}.$$

Thus both the domain and the range of σ_n are identified with $[n] := \{1, \dots, n\}$. This is the exponential-race representation of the Luce ranking model; see Luce [3].

For a permutation σ, let $C_r(\sigma)$ denote its number of r-cycles, let $L_\sigma(v)$ be the length of the cycle containing v, and write

$$L_n(v) := L_{\sigma_n}(v), \qquad C_{n,r} := C_r(\sigma_n).$$

Let I_n be uniform on $[n]$, independent of the clocks. For two probability measures on a finite or countable space we use the convention

$$\|\mu - \nu\|_{\mathrm{TV}} := \sup_A |\mu(A) - \nu(A)| = \frac{1}{2} \sum_x |\mu(x) - \nu(x)|.$$

We write $C_c^+((0,\infty))$ for the nonnegative continuous functions with compact support in $(0,\infty)$.

The endpoint construction needs only a localized bound on the probability that a uniform root lies in a short cycle. This bound is proved internally below from the exact master switch and selected-factor moment bounds. Corollary 4.3 gives

$$\mathbb{P}(L_n(I_n) \le a) \le C_\kappa a/n \qquad (a \le n/2).$$

Consequently no macroscopic PD(1) or GEM(1) input is used in this paper.

Theorem 1.1 (Conjecture 13.1)**.** *Assume* (1).

(i) *For every pair of integer sequences $r_-(n), r_+(n)$ satisfying*

$$r_-(n) \longrightarrow \infty, \qquad r_+(n) = o(n), \qquad r_-(n) \le r_+(n),$$

one has

$$\sup_{r_-(n) \le r \le r_+(n)} |r\mathbb{E}C_{n,r} - 1| \longrightarrow 0. \tag{2}$$

In particular, $\mathbb{E}C_{n,r_n} \sim 1/r_n$ whenever $r_n \to \infty$ and $r_n = o(n)$.

(ii) *If $a_n \to \infty$ and $a_n = o(n)$, then*

$$\Xi_n := \sum_{C \in \mathrm{Cyc}(\sigma_n)} \delta_{|C|/a_n} \tag{3}$$

converges vaguely in distribution on $(0, \infty)$ to

$$\mathrm{PPP}\left(\frac{\mathrm{d}x}{x}\right).$$

Equivalently, cycle counts in finitely many pairwise disjoint compact intervals converge jointly to independent Poisson random variables with means given by the $\mathrm{d}x/x$-masses of those intervals.

Remark 1.2. The endpoint theorem proved below is stronger than the scalar identity needed for the main result: it treats every fixed number of roots and controls the endpoint vector in mean conditional total variation. Only the one- and two-root cases are used to prove L^2 concentration of the scalar switching intensity. Part (ii) is then obtained from the same one-cycle identity under a bounded downward exponential tilt.

2 An elementary orbit-intersection bound

Lemma 2.1 (Intersection of two orbit segments)**.** *Let σ be any permutation of $[n]$, and let I, J be independent uniform elements of $[n]$. For integers $r, s \geq 0$,*

$$\mathbb{P}\left(\{\sigma^t(I) : 0 \leq t \leq r\} \cap \{\sigma^u(J) : 0 \leq u \leq s\} \neq \varnothing \;\middle|\; \sigma, I\right) \leq \frac{r+s+1}{n}. \tag{4}$$

Proof. Put $A = \{\sigma^t(I) : 0 \leq t \leq r\}$. The set of starting points x whose first s iterates intersect A is

$$\bigcup_{u=0}^{s} \sigma^{-u}(A).$$

All these points lie on the cycle of I. Along that cycle, A is a contiguous arc, possibly with repetitions if the cycle is short, and the union of its first s backward shifts has at most $r+s+1$ distinct points. Since J is uniform, (4) follows. □

3 An exact clock-swap identity

For the remainder of the paper we suppress the subscript n when no confusion is possible. For all labels i, j, define

$$R_{ij}(X) := \exp\{-(\theta_i - \theta_j)(X_j - X_i)\}, \tag{5}$$

so in particular $R_{ii} = 1$. For distinct labels i, j, let T_{ij} be the involution that exchanges X_i and X_j and leaves all other clocks unchanged.

Lemma 3.1 (Change of measure under a clock swap)**.** *For distinct i, j,*

$$\sigma_{T_{ij}X} = \sigma_X \circ (i\, j), \tag{6}$$

and

$$\frac{\mathrm{d}(\mathbb{P}\circ T_{ij})}{\mathrm{d}\mathbb{P}}(X) = R_{ij}(X). \tag{7}$$

Consequently, for every nonnegative measurable G,

$$\mathbb{E}\big[R_{ij}(X)G(T_{ij}X)\big] = \mathbb{E}G(X). \tag{8}$$

Proof. Swapping X_i and X_j exchanges the ranks assigned to the labels i and j, proving (6). The joint density of X is

$$f(x) = \prod_{k=1}^{n} \theta_k e^{-\theta_k x_k}\mathbf{1}_{\{x_k>0\}}.$$

Thus

$$\frac{f(T_{ij}x)}{f(x)} = \exp\{(\theta_i-\theta_j)(x_i-x_j)\} = R_{ij}(x),$$

which proves (7); (8) is the associated change-of-variables identity. □

The following marked identity is the central exact relation. For a permutation σ and a label j, write $\mathcal{C}_\sigma(j)$ for the vertex set of the cycle containing j.

Proposition 3.2 (Marked master switching identity)**.** *Let $1 \le r < n$, and let $\Psi : [n]^2 \times S_n \to [0,\infty)$ be arbitrary. Then*

$$\begin{aligned}\mathbb{E}\sum_{\substack{i\in[n]\\ L_\sigma(i)>r}} R_{i,\sigma^r(i)}(X)\Psi\big(i,\sigma^r(i),\sigma\circ(i\,\sigma^r(i))\big)\\ = \mathbb{E}\sum_{\substack{j\in[n]\\ L_\sigma(j)=r}}\ \sum_{i\notin\mathcal{C}_\sigma(j)} \Psi(i,j,\sigma).\end{aligned} \tag{9}$$

In particular, for every $\Phi : S_n \to [0,\infty)$,

$$r(n-r)\mathbb{E}\big[C_r(\sigma)\Phi(\sigma)\big] = \mathbb{E}\sum_{\substack{i\in[n]\\ L_\sigma(i)>r}} R_{i,\sigma^r(i)}(X)\Phi\big(\sigma\circ(i\,\sigma^r(i))\big). \tag{10}$$

Both identities extend by linearity whenever the displayed terms are integrable.

Proof. For distinct i,j, let

$$A_{ij}^{(r)} = \{j = \sigma^r(i),\ L_\sigma(i) > r\}.$$

On $A_{ij}^{(r)}$, the labels i,j lie on the same cycle, at directed distance r. Right-composition by $(i\,j)$ swaps the outgoing edges of i and j and splits that cycle into cycles of lengths r and $L_\sigma(i)-r$; the r-cycle contains j.

By Lemma 3.1,

$$\begin{aligned}\mathbb{E}\left[\mathbf{1}_{A_{ij}^{(r)}}R_{ij}(X)\Psi\big(i,j,\sigma\circ(i\,j)\big)\right]\\ = \mathbb{E}\left[\mathbf{1}_{A_{ij}^{(r)}}(T_{ij}X)\Psi(i,j,\sigma_X)\right].\end{aligned}$$

In terms of σ_X, the event $A_{ij}^{(r)}(T_{ij}X)$ says exactly that j belongs to an r-cycle and that i lies outside $\mathcal{C}_{\sigma_X}(j)$: right-composition by $(i\,j)$ then merges the two cycles, and j is at directed distance r from i in the merged cycle. Summing over ordered pairs $i \neq j$ proves (9).

For $\Psi(i,j,\sigma) = \Phi(\sigma)$, every r-cycle contributes r choices for j, and for each such j there are $n-r$ choices for i outside that cycle. This gives (10). □

Define the switching intensity

$$S_{n,r} := \sum_{i:L_n(i)>r} R_{i,\sigma_n^r(i)}(X). \tag{11}$$

Taking $\Phi \equiv 1$ in Proposition 3.2 gives the exact identity

$$r(n-r)\mathbb{E}C_{n,r} = \mathbb{E}S_{n,r}. \tag{12}$$

Thus Theorem 1.1(i) will follow once we prove $S_{n,r}/n \to 1$ in L^1, uniformly over mesoscopic windows.

4 Uniform integrability of the swap factors

The factors R_{ij} are unbounded. Comparability of the rates gives precisely the uniform integrability needed below.

Set

$$\lambda_\kappa := \begin{cases} \infty, & \kappa = 1, \\ \dfrac{\kappa}{\kappa-1}, & \kappa > 1. \end{cases} \tag{13}$$

Lemma 4.1 (Selected-pair tail bound)**.** *Suppose $\kappa > 1$. For every n, r and $u \geq 0$,*

$$\frac{1}{n}\mathbb{E}\sum_{i:L_n(i)>r} \mathbf{1}_{\{\log R_{i,\sigma_n^r(i)} > u\}} \leq 2e^{-\lambda_\kappa u}. \tag{14}$$

Consequently, for every $A \geq 1$,

$$\frac{1}{n}\mathbb{E}\sum_{i:L_n(i)>r} R_{i,\sigma_n^r(i)}\mathbf{1}_{\{R_{i,\sigma_n^r(i)} > A\}} \leq C_\kappa A^{-1/(\kappa-1)}. \tag{15}$$

If $\kappa = 1$, then $R_{ij} \equiv 1$.

Proof. Write $j = \sigma_n^r(i)$. If $\theta_i > \theta_j$ and $\log R_{ij} > u$, then

$$(\theta_i - \theta_j)(X_i - X_j) > u,$$

so

$$X_i > \frac{u}{\theta_i - \theta_j} \geq \frac{u}{\theta_i - 1}.$$

For $\theta_i > 1$,

$$\mathbb{P}\left(X_i > \frac{u}{\theta_i - 1}\right) = \exp\left\{-\frac{\theta_i}{\theta_i - 1}u\right\} \leq e^{-\lambda_\kappa u},$$

because $a/(a-1)$ is decreasing on $(1, \kappa]$. Summing over i gives at most $ne^{-\lambda_\kappa u}$.

If $\theta_j > \theta_i$ and $\log R_{ij} > u$, the same argument gives $X_j > u/(\theta_j - 1)$. The set $\{i : L_n(i) > r\}$ is invariant under σ_n^r, and the map $i \mapsto \sigma_n^r(i)$ is a bijection of this set. Reindexing by j therefore gives another contribution at most $ne^{-\lambda_\kappa u}$. This proves (14).

For a nonnegative random variable Z,

$$\mathbb{E}[Z\mathbf{1}_{\{Z>A\}}] = A\mathbb{P}(Z > A) + \int_A^\infty \mathbb{P}(Z > t)\,\mathrm{d}t.$$

Apply this layer-cake identity to the averaged counting measure in (14), using $u = \log t$. Since $t^{-\lambda_\kappa}$ is integrable at infinity,

$$A^{1-\lambda_\kappa} + \int_A^\infty t^{-\lambda_\kappa}\,\mathrm{d}t \le C_\kappa A^{1-\lambda_\kappa} = C_\kappa A^{-1/(\kappa-1)}.$$

This proves (15). □

Corollary 4.2 (Moments of selected swap factors)**.** *For every $1 \le p < \lambda_\kappa$,*

$$\sup_{n\ge 1} \sup_{1\le r<n} \frac{1}{n}\mathbb{E} \sum_{i:L_n(i)>r} R_{i,\sigma_n^r(i)}^p < \infty. \tag{16}$$

Moreover,

$$\sup_{n\ge 1} \max_{i,j\in[n]} \mathbb{E}R_{ij}^p < \infty. \tag{17}$$

Proof. When $\kappa = 1$, every swap factor equals one. Suppose $\kappa > 1$. For $t \ge 1$, (14) with $u = \log t$ gives

$$\frac{1}{n}\mathbb{E} \sum_{i:L_n(i)>r} \mathbf{1}_{\{R_{i,\sigma_n^r(i)}>t\}} \le 2t^{-\lambda_\kappa}.$$

Layer-cake integration for the pth moment is finite uniformly in n, r because $p < \lambda_\kappa$. Repeating the same two-case argument for a fixed ordered pair gives $\mathbb{P}(\log R_{ij} > u) \le 2e^{-\lambda_\kappa u}$ uniformly in i, j. A second layer-cake integration proves (17). □

Corollary 4.3 (Crude cycle and root bounds)**.** *Uniformly in $1 \le r < n$,*

$$\mathbb{E}C_{n,r} \le \frac{C_\kappa n}{r(n-r)}. \tag{18}$$

If I_n is uniform on $[n]$, independently of the clocks, then for $a \le n/2$,

$$\mathbb{P}\big(L_n(I_n) \le a\big) \le C_\kappa a/n. \tag{19}$$

In particular, submacroscopic root-cycle escape follows directly from uniform rate comparability.

Proof. Take $p = 1$ in Corollary 4.2. Then $\mathbb{E}S_{n,r} \le C_\kappa n$, and the exact identity (12) gives (18). Since

$$\mathbb{P}\big(L_n(I_n) = r\big) = \frac{r\mathbb{E}C_{n,r}}{n},$$

we obtain, for $a \le n/2$,

$$\mathbb{P}\big(L_n(I_n) \le a\big) \le C_\kappa \sum_{r\le a} \frac{1}{n-r} \le C'_\kappa a/n.$$

□

For $A \geq 1$, define

$$F_A(i,j) := R_{ij} \wedge A. \tag{20}$$

Lemma 4.4 (Independent-pair normalization)**.** *For each fixed ordered pair i, j,*

$$\mathbb{E}R_{ij} = 1. \tag{21}$$

If $\kappa > 1$, then uniformly in i, j and $A \geq 1$,

$$\mathbb{E}\left[R_{ij}\mathbf{1}_{\{R_{ij}>A\}}\right] \leq C_\kappa A^{-1/(\kappa-1)}. \tag{22}$$

If I, J are independent uniform labels, independent of the clocks, then

$$\mu_{n,A} := \mathbb{E}F_A(I,J) \tag{23}$$

satisfies

$$0 \leq 1 - \mu_{n,A} \leq C_\kappa A^{-1/(\kappa-1)} \tag{24}$$

when $\kappa > 1$, uniformly in n; for $\kappa = 1$ one has $\mu_{n,A} = 1$ for $A \geq 1$.

Proof. If $i = j$, then $R_{ii} = 1$. If $i \neq j$, independence gives

$$\begin{aligned}\mathbb{E}R_{ij} &= \mathbb{E}e^{(\theta_i-\theta_j)X_i}\mathbb{E}e^{-(\theta_i-\theta_j)X_j} \\ &= \frac{\theta_i}{\theta_j}\frac{\theta_j}{\theta_i} = 1.\end{aligned}$$

The moment-generating factors are finite because the positive exponent is always strictly smaller than the corresponding exponential rate. For a fixed pair, the same two-case argument as in Lemma 4.1 gives $\mathbb{P}(\log R_{ij} > u) \leq 2e^{-\lambda_\kappa u}$, uniformly in i, j. Layer-cake integration therefore proves (22). Since

$$1 - \mathbb{E}(R_{IJ} \wedge A) = \mathbb{E}[(R_{IJ} - A)_+] \leq \mathbb{E}[R_{IJ}\mathbf{1}_{\{R_{IJ}>A\}}],$$

averaging over I, J proves (24). □

5 Conditional matching structure inside clock blocks

We next record, with proof, the only structural fact about the Luce permutation used in the endpoint argument.

Let

$$0 = t_0 < t_1 < \cdots < t_d = T$$

be a deterministic partition, and let $B_a = (t_{a-1}, t_a]$. Put

$$H_a = \{i : X_i \in B_a\}, \qquad m_a = |H_a|,$$

and let $H_* = \{i : X_i > T\}$. The ranks occupied by clocks in B_a form the consecutive interval

$$Q_a = \left\{1 + \sum_{c<a} m_c, \ldots, \sum_{c\leq a} m_c\right\},$$

while Q_* is the terminal rank interval of size $|H_*|$.

Let $\mathcal{H}$ be generated by the sets H_a, H_* and, for every block, the ordered but unlabeled clock values in that block. Thus $\mathcal{H}$ does not reveal which label in H_a receives which rank in Q_a.

Lemma 5.1 (Conditional permanental rows)**.** *Conditional on $\mathcal{H}$, the restrictions*

$$\sigma : H_a \longrightarrow Q_a$$

are independent over the blocks. If the ordered clock values in block a are $s_{a,1} < \cdots < s_{a,m_a}$, then for a bijection $\pi : H_a \to Q_a$,

$$\mathbb{P}(\sigma|_{H_a} = \pi \mid \mathcal{H}) = \frac{1}{Z_a} \prod_{i \in H_a} e^{-\theta_i s_{a,\pi(i)-\min Q_a+1}}, \tag{25}$$

where Z_a is the corresponding permanent. After any compatible collection of source-target pairs has been exposed, the same statement holds on each residual submatrix, independently across rows.

Proof. The joint density of the clocks in block a, before labels are assigned to the ordered values, is

$$\prod_{i \in H_a} \theta_i e^{-\theta_i x_i}.$$

For a fixed assignment of the ordered values to the labels, the factor $\prod_{i \in H_a} \theta_i$ is independent of the assignment, leaving the weight in (25). Since the original clock density factors over labels and the blocks are disjoint, the assignments in different blocks are conditionally independent. Conditioning further on a set of compatible assignments simply fixes the corresponding factors and leaves the same weighted-bijection law on the residual source and target sets. □

Lemma 5.2 (Hereditary flatness)**.** *Suppose B_a has width at most h. Conditional on $\mathcal{H}$ and on any compatible previous exposures, let a residual row have $m \geq 1$ targets. For every residual source x and residual target y,*

$$\frac{e^{-\beta}}{m} \leq \mathbb{P}(\sigma(x) = y \mid \mathcal{H}, \textit{exposures}) \leq \frac{e^{\beta}}{m}, \qquad \beta := \kappa h. \tag{26}$$

Consequently, the one-edge law is at total-variation distance at most $\frac{1}{2}(e^{\beta} - 1)$ from the uniform law on the residual targets. The same bound holds after pushing the targets forward through any partition into categories.

Proof. Fix two residual targets y, z. In every residual matching with $x \mapsto y$, let u be the source assigned to z, and interchange the targets assigned to x and u. This is a bijection onto the residual matchings with $x \mapsto z$. If the corresponding clock values are $s_y, s_z \in B_a$, the ratio of the two matching weights is

$$\exp\{(\theta_x - \theta_u)(s_y - s_z)\},$$

which lies in $[e^{-\kappa h}, e^{\kappa h}]$. Summing over matchings shows that the marginal weights for any two targets have ratio in the same interval. If p_y are m positive numbers summing to one and $p_y/p_z \in [e^{-\beta}, e^{\beta}]$ for all y, z, then $e^{-\beta}/m \leq p_y \leq e^{\beta}/m$, proving (26). The total-variation assertion follows from

$$\frac{1}{2} \sum_y \left| p_y - \frac{1}{m} \right| \leq \frac{1}{2}(e^{\beta} - 1),$$

and total variation cannot increase under a measurable map. □

6 A core-tail environment at the scale of an endpoint window

Fix an integer $q \geq 1$ and integer sequences $r_-(n), r_+(n)$ with

$$r_- \longrightarrow \infty, \qquad r_+ = o(n), \qquad r_- \leq r_+.$$

The endpoint argument uses the cycle-length information only through

$$\zeta_n := \mathbb{P}(L_n(I_n) \leq r_+). \tag{27}$$

By Corollary 4.3, for all sufficiently large n,

$$\zeta_n \leq C_\kappa r_+/n. \tag{28}$$

Set

$$\begin{aligned} K &:= qr_+, & \delta &:= \frac{K}{n}, & L_\delta &:= \log\frac{1}{\delta}, \\ h &:= \delta^{1/4}, & T &:= \frac{1}{4\kappa}L_\delta, & \rho &:= e^{-T} = \delta^{1/(4\kappa)}, \\ M &:= n\delta^{1/2} = \sqrt{nK}. \end{aligned} \tag{29}$$

All these quantities may depend on n; q is fixed. We work only for sufficiently large n, so $0 < \delta < 1/2$.

Partition $[0, T]$ into $d = \lceil T/h \rceil$ equal intervals. Their common width $\ell = T/d$ satisfies $h/2 \leq \ell \leq h$ for all large n. Fix a constant $\tau_0 \in (0, 1)$, and call a finite block *early* if its right endpoint is at most τ_0. Let E be the set of early block indices, and put $d_{\mathrm{E}} = |\mathsf{E}|$. Then

$$d \leq Ch^{-1}L_\delta, \qquad ch^{-1} \leq d_{\mathrm{E}} \leq Ch^{-1}. \tag{30}$$

For blocks $a, b \in \{1, \ldots, d, *\}$, define

$$N_{ab} := |Q_a \cap H_b|. \tag{31}$$

We use the following elementary Bernoulli bounds. If Z_j are independent Bernoulli variables and $\mu = \sum_j \mathbb{E}Z_j$, then

$$\mathbb{P}\left(\sum_j Z_j \leq \frac{\mu}{2}\right) \leq e^{-\mu/8}, \tag{32}$$

and, for $x \geq 0$,

$$\mathbb{P}\left(\sum_j Z_j \geq \mathrm{e}\mu + x\right) \leq e^{-x}. \tag{33}$$

The first is the usual multiplicative Chernoff bound. For the second, $\mathbb{E}e^{\sum Z_j} \leq \exp((\mathrm{e}-1)\mu)$ and Markov's inequality give

$$\mathbb{P}\left(\sum Z_j \geq \mathrm{e}\mu + x\right) \leq e^{(\mathrm{e}-1)\mu - \mathrm{e}\mu - x} \leq e^{-x}.$$

Lemma 6.1 (Uniform core-tail environment)**.** *There are constants $c, C > 0$, depending only on κ and τ_0, and events $\mathcal{G}_n$ such that*

$$\mathbb{P}(\mathcal{G}_n^c) \leq C\left[h^{-1}L_\delta e^{-cM} + n^2h^{-1}e^{-chM} + n^{-2}\right]. \tag{34}$$

In particular, $\mathbb{P}(\mathcal{G}_n) \to 1$. On $\mathcal{G}_n$:

(i) *Every finite block satisfies*

$$m_a \geq cM. \tag{35}$$

(ii) *For every integer interval $J \subset [n]$ with $|J| \geq cM$ and every early block $b \in \mathsf{E}$,*

$$|J \cap H_b| \geq ch|J|. \tag{36}$$

(iii) *For every integer interval $J \subset [n]$ with $|J| \geq cM$,*

$$|J \cap H_*| \leq C(\rho|J| + \log n). \tag{37}$$

Moreover,

$$m_* \leq Cn\rho. \tag{38}$$

(iv) *Consequently, for every finite source row a,*

$$N_{ab} \geq chm_a \qquad (b \in \mathsf{E}), \tag{39}$$

$$N_{a*} \leq C(\rho m_a + \log n). \tag{40}$$

Proof. For a finite block $B_a = (u, u + \ell]$ and any label i,

$$\begin{aligned}\mathbb{P}(X_i \in B_a) &= e^{-\theta_i u}(1 - e^{-\theta_i \ell}) \\ &\geq e^{-\kappa T}(1 - e^{-\ell}) \geq c\delta^{1/4}h = c\delta^{1/2}.\end{aligned}$$

Thus $\mathbb{E}m_a \geq cM$, and (32), followed by a union bound over d, proves (35) with failure probability at most $de^{-cM} = o(1)$.

If b is early, then uniformly in i,

$$\mathbb{P}(X_i \in B_b) \geq e^{-\kappa\tau_0}(1 - e^{-\ell}) \geq ch.$$

For a fixed interval J with $|J| \geq cM$, the expected value of $|J \cap H_b|$ is at least $ch|J|$. The lower Chernoff bound and a union bound over at most $n^2 d_{\mathrm{E}}$ pairs (J, b) prove (36), because

$$hM = n\delta^{3/4} = n^{1/4}K^{3/4} \gg \log n. \tag{41}$$

Here $K \geq r_- \to \infty$.

For the tail, $\mathbb{P}(X_i > T) = e^{-\theta_i T} \leq e^{-T} = \rho$. For a fixed J, (33) with $\mu \leq \rho|J|$ and $x = 4\log n$ gives

$$\mathbb{P}(|J \cap H_*| \geq \mathrm{e}\rho|J| + 4\log n) \leq n^{-4}.$$

A union bound over at most n^2 intervals gives (37). Applying the same estimate to $J = [n]$ proves (38), since $n\rho \gg \log n$ under (29).

Finally, each Q_a is an integer interval of length $m_a \geq cM$. Applying (36) and (37) to $J = Q_a$ gives (39) and (40). The three union bounds above, together with (30), give (34). □

7 Finite-endpoint mixing relative to the coarse environment

Let $U_n := \mathrm{Unif}([n])$. Let $I_1, \dots, I_q$ be independent U_n labels, independent of the clocks, and for $\mathbf{r} = (r_1, \dots, r_q) \in [r_-, r_+]^q$ put

$$E_\ell(\mathbf{r}) := \sigma_n^{r_\ell}(I_\ell), \qquad 1 \le \ell \le q. \tag{42}$$

Definition 7.1 (Admissible terminal depth)**.** A deterministic integer sequence $m = m_n$ is *admissible* for the window $[r_-, r_+]$ if, for all sufficiently large n,

$$1 \le m \le \frac{r_-}{3}, \tag{43}$$

and

$$m \longrightarrow \infty, \qquad mh \longrightarrow 0, \qquad m\rho \longrightarrow 0, \qquad \frac{m^2}{M} \longrightarrow 0, \qquad \frac{m \log n}{M} \longrightarrow 0, \qquad \frac{m}{hM} \longrightarrow 0. \tag{44}$$

Proposition 7.2 (Free-depth finite-endpoint mixing in mean conditional total variation)**.** *Let m be any admissible terminal depth. For every fixed $q \ge 1$, with the coarse block sigma-field $\mathcal{H}$ constructed in Section 6 using $K = qr_+$,*

$$\Delta_{n,q} := \sup_{\mathbf{r} \in [r_-, r_+]^q} \mathbb{E}\left[\left\| \mathcal{L}\left((E_1(\mathbf{r}), \dots, E_q(\mathbf{r})) \mid \mathcal{H}, I_1, \dots, I_q \right) - U_n^{\otimes q} \right\|_{\mathrm{TV}} \right] \longrightarrow 0. \tag{45}$$

More quantitatively, there are constants $c, C > 0$, depending only on κ, q and τ_0, such that

$$\Delta_{n,q} \le C \Bigg[(1-\alpha)^{m-1} + mh + \frac{m^2}{M} + m\rho + \frac{m \log n}{M} + \rho + \delta + \delta^{1/2} L_\delta + h^{-1} L_\delta e^{-cM} + n^2 h^{-1} e^{-chM} + n^{-2} \Bigg]. \tag{46}$$

Here $\alpha > 0$ is the Doeblin constant in Lemma 7.8.

Corollary 7.3 (Square-root-depth specialization)**.** *The choice*

$$m = \left\lfloor \min\left\{ \frac{r_-}{3}, \sqrt{L_\delta} \right\} \right\rfloor \tag{47}$$

is admissible and therefore recovers the qualitative endpoint theorem.

Proof. Both entries in the minimum tend to infinity. Since $m \le \sqrt{L_\delta}$ and $x^c \sqrt{\log(1/x)} \to 0$ as $x \downarrow 0$ for every $c > 0$, one has $mh \to 0$ and $m\rho \to 0$. Also $M = \sqrt{nK} \ge \sqrt{nr_-}$ and $m \le \sqrt{\log n}$, so

$$\frac{m^2}{M} \le \frac{\log n}{\sqrt{nr_-}} \to 0, \qquad \frac{m \log n}{M} \le \frac{(\log n)^{3/2}}{\sqrt{nr_-}} \to 0.$$

Finally, $hM = n\delta^{3/4} = n^{1/4} K^{3/4}$, whence $m/(hM) \le \sqrt{\log n}/(n^{1/4} K^{3/4}) \to 0$. □

Remark 7.4 (Conditioning level). Since the roots are independent of $\mathcal{H}$ and have law $U_n^{\otimes q}$, Proposition 7.2 is equivalently the statement

$$\sup_{\mathbf{r}\in[r_-,r_+]^q} \mathbb{E}\left\|\mathcal{L}\left((I_1,\dots,I_q,E_1(\mathbf{r}),\dots,E_q(\mathbf{r}))\mid \mathcal{H}\right) - U_n^{\otimes 2q}\right\|_{\mathrm{TV}} \longrightarrow 0.$$

The estimate is conditional only on the coarse environment, in mean over that environment. It is not a quenched statement conditional on all exact clocks, under which every endpoint is deterministic.

The proof occupies the rest of the section. Fix an arbitrary admissible terminal depth m. The bound $m \le r_-/3$ makes every prefix length $k_\ell = r_\ell - m$ nonnegative and leaves at least two thirds of each observed orbit in the frozen prefix. We use repeatedly the admissibility condition

$$\frac{m}{hM} \longrightarrow 0. \tag{48}$$

Since $h \le 1$, this also implies $m/M \to 0$.

7.1 Prefix depletion is negligible

For each root I_ℓ, write $v_{\ell,t} = \sigma_n^t(I_\ell)$ and put $k_\ell = r_\ell - m$. The following counts involve only the prefixes of length k_ℓ:

$$V_a := \sum_{\ell=1}^{q}\sum_{t=0}^{k_\ell-1} \mathbf{1}_{\{v_{\ell,t}\in H_a\}},$$
$$U_{ab} := \sum_{\ell=1}^{q}\sum_{t=0}^{k_\ell-1} \mathbf{1}_{\{v_{\ell,t}\in H_a,\ v_{\ell,t+1}\in H_b\}}.$$

Define the prefix-depletion event

$$\mathcal{D}_{n,\mathbf{r}} := \left\{V_a \le \frac{m_a}{4}\ \forall a \le d\right\} \cap \left\{U_{ab} \le \frac{N_{ab}}{2}\ \ \forall a\le d,\ b\in \mathsf{E}\right\}. \tag{49}$$

Lemma 7.5 (The prefix-depletion event is likely)**.** *Uniformly in $r_\ell \in [r_-, r_+]$,*

$$\mathbb{P}(\mathcal{D}_{n,\mathbf{r}}^c) \le C_q\left[\delta^{3/4}L_\delta + \delta^{1/2}L_\delta\right] = o(1). \tag{50}$$

Proof. Conditional on the full permutation, $v_{\ell,t} = \sigma_n^t(I_\ell)$ is uniform on $[n]$ for every fixed (ℓ,t). Hence

$$\mathbb{E}(V_a \mid \sigma_n, X) = \frac{(\sum_\ell k_\ell)m_a}{n} \le \frac{Km_a}{n}.$$

Markov's inequality gives

$$\mathbb{P}(V_a > m_a/4 \mid \sigma_n, X) \le 4K/n = 4\delta.$$

Similarly,

$$\#\{x\in H_a : \sigma_n(x)\in H_b\} = |Q_a \cap H_b| = N_{ab},$$

so

$$\mathbb{E}(U_{ab}\mid \sigma_n, X) \le \frac{KN_{ab}}{n}, \qquad \mathbb{P}(U_{ab} > N_{ab}/2 \mid \sigma_n, X) \le 2\delta.$$

A union bound, together with (30), gives

$$\mathbb{P}(\mathcal{D}_{n,\mathbf{r}}^c \mid \sigma_n, X) \le 4d\delta + 2dd_{\mathrm{E}}\delta \le C\left(\delta^{3/4}L_\delta + \delta^{1/2}L_\delta\right) = o(1).$$

□

7.2 The frozen residual rows

Expose, for every ℓ, the prefix

$$v_{\ell,0} \to v_{\ell,1} \to \cdots \to v_{\ell,r_\ell - m}.$$

We declare failure if a path closes or if two exposed path segments intersect. By (27), the closure probability is at most $q\zeta_n$. Lemma 2.1 and a union bound over pairs give an intersection probability at most $Cq^2 r_+/n$. Thus the total failure probability is $o(1)$, uniformly in the chosen lengths. On its complement the exposed prefixes are vertex-disjoint directed paths.

For every block a, let $S_a^0 \subset H_a$ be the unexposed sources and $T_a^0 \subset Q_a$ the unused targets after the prefixes have been exposed. Every exposed edge from a source in H_a uses a target in Q_a, so

$$|S_a^0| = |T_a^0| =: m_a^0. \tag{51}$$

Put $S^0 = \bigcup_a S_a^0$, $T^0 = \bigcup_a T_a^0$, and $N_0 = |S^0| = |T^0|$. Since the prefixes are disjoint open paths,

$$T^0 \setminus S^0 = \{I_1, \dots, I_q\}, \qquad S^0 \setminus T^0 = \{v_{1,r_1-m}, \dots, v_{q,r_q-m}\}. \tag{52}$$

In particular both differences have cardinality q.

Define residual cells

$$N_{ab}^0 := |T_a^0 \cap S_b^0|. \tag{53}$$

Lemma 7.6 (Residual core estimates)**.** *On $\mathcal{G}_n \cap \mathcal{D}_{n,\mathbf{r}}$, and on the event that the prefixes are disjoint, for all sufficiently large n:*

$$m_a^0 \ge \frac{3}{4} m_a \ge cM \qquad (a \le d), \tag{54}$$

$$N_{ab}^0 \ge chm_a^0 \qquad (a \le d,\ b \in \mathsf{E}), \tag{55}$$

$$\frac{N_{a*}^0}{m_a^0} \le C\left(\rho + \frac{\log n}{M}\right) \qquad (a \le d), \tag{56}$$

$$p_*^0 := \frac{m_*^0}{N_0} \le C\rho. \tag{57}$$

Proof. The number of prefix sources removed from row a is at most V_a, so (54) follows from (49) and Lemma 6.1.

A target deleted from the initial cell $Q_a \cap H_b$ is counted by U_{ab}. Passing from H_b to the residual source set S_b^0 can remove additional labels from $T_a^0 \cap H_b$ only when a label has been used as a source but not as a target. By (52) there are only q such labels. Hence

$$N_{ab}^0 \ge N_{ab} - U_{ab} - q \ge \frac{N_{ab}}{2} - q.$$

On $\mathcal{G}_n$, $N_{ab} \ge chm_a \ge chM$, and $hM \to \infty$ by (41). Thus, after changing c, this proves (55). The tail intersection can only decrease, so $N_{a*}^0 \le N_{a*}$; combine (40) with (54) to get (56). Finally $m_*^0 \le m_* \le Cn\rho$ and $N_0 = n - O(K) = n(1 - o(1))$, proving (57). □

7.3 The reset kernel

We first isolate the deterministic comparison mechanism used below.

Lemma 7.7 (Adaptive Doeblin comparison)**.** *Let P be a Markov kernel on a finite or countable state space such that, for some $\alpha \in (0,1]$,*

$$\|\mu P - \mu' P\|_{\mathrm{TV}} \le (1-\alpha)\|\mu - \mu'\|_{\mathrm{TV}} \tag{58}$$

for all probability measures μ, μ'. If p satisfies $\|pP - p\|_{\mathrm{TV}} \le \zeta$, then for every probability measure μ and $j \ge 0$,

$$\|\mu P^j - p\|_{\mathrm{TV}} \le (1-\alpha)^j \|\mu - p\|_{\mathrm{TV}} + \frac{\zeta}{\alpha}. \tag{59}$$

Suppose in addition that an adapted process $(Y_t)_{t=0}^j$ can be coupled to a P-chain $(Z_t)_{t=0}^j$ with the same initial law so that

$$\mathbb{P}(Y_t \ne Z_t \text{ for some } t \le j) \le \beta_j.$$

Then

$$\|\mathcal{L}(Y_j) - p\|_{\mathrm{TV}} \le \beta_j + (1-\alpha)^j + \frac{\zeta}{\alpha}. \tag{60}$$

If, before the first disagreement, the conditional probability of a new disagreement at transition t is at most η_t, then one may take $\beta_j \le \sum_{t=0}^{j-1} \eta_t$.

Proof. Writing $d_j = \|pP^j - p\|_{\mathrm{TV}}$, (58) gives $d_{j+1} \le (1-\alpha)d_j + \zeta$ and $d_0 = 0$. Hence $d_j \le \zeta/\alpha$. Combining this with

$$\|\mu P^j - pP^j\|_{\mathrm{TV}} \le (1-\alpha)^j \|\mu - p\|_{\mathrm{TV}}$$

proves (59). The coupling inequality gives $\|\mathcal{L}(Y_j) - \mathcal{L}(Z_j)\|_{\mathrm{TV}} \le \beta_j$, which proves (60). The final assertion follows by a union bound over the successive conditional coupling failures. □

Let the state space be $\mathcal{S} = \{1, \dots, d, *\}$. Let ν be the uniform probability measure on the early states E. If $m_a^0 > 0$, put

$$d_a^0 := \frac{|T_a^0 \setminus S^0|}{m_a^0}$$

and define

$$\widehat{P}_{ab}^0 := \frac{N_{ab}^0}{m_a^0} + d_a^0 \nu_b. \tag{61}$$

The matrix $(N_{ab}^0/m_a^0)_b$ is substochastic: the missing targets are exactly the roots lying in the rank row T_a^0, and they cannot serve as new sources. Thus (61) is stochastic. If $m_a^0 = 0$, set $\widehat{P}^0(a,\cdot) = \nu(\cdot)$. Finally define the *reset kernel* $\overline{P}^0$ by

$$\overline{P}^0(a,\cdot) = \begin{cases} \widehat{P}^0(a,\cdot), & a \le d, \\ \nu(\cdot), & a = *. \end{cases} \tag{62}$$

Let

$$p_a^0 := \frac{m_a^0}{N_0}. \tag{63}$$

For later use, let $\lambda_a^0 = \mathrm{Unif}(T_a^0)$ when $m_a^0 > 0$; if $m_a^0 = 0$, choose λ_a^0 arbitrarily.

Lemma 7.8 (Doeblin minorization and approximate invariance)**.** *On the event of Lemma 7.6, there is a constant $\alpha = \alpha(\kappa, \tau_0) > 0$ such that*

$$\overline{P}^0(a, \cdot) \geq \alpha\nu(\cdot) \qquad (a \in \mathcal{S}). \tag{64}$$

Moreover,

$$\|p^0\overline{P}^0 - p^0\|_{\mathrm{TV}} \leq C\left(\rho + \frac{q}{N_0}\right). \tag{65}$$

Consequently, for every state a and every $j \geq 0$,

$$\|\delta_a(\overline{P}^0)^j - p^0\|_{\mathrm{TV}} \leq (1-\alpha)^j + C\left(\rho + \frac{q}{N_0}\right). \tag{66}$$

Proof. For finite a and early b, (55) gives $\widehat{P}^0_{ab} \geq ch$. Since $d_{\mathrm{E}} = |\mathsf{E}| \geq c/h$, there is a constant $\alpha > 0$ for which $ch \geq \alpha/d_{\mathrm{E}} = \alpha\nu_b$. The tail row equals ν, so (64) follows.

Let

$$r_a = |T^0_a \setminus S^0|, \qquad e_b = |S^0_b \setminus T^0|.$$

By (52), $\sum_a r_a = \sum_b e_b = q$. For every b,

$$\begin{aligned}(p^0\widehat{P}^0)_b &= \frac{1}{N_0}\sum_a N^0_{ab} + \frac{1}{N_0}\sum_a r_a\nu_b \\ &= \frac{|S^0_b \cap T^0|}{N_0} + \frac{q}{N_0}\nu_b,\end{aligned}$$

whereas

$$p^0_b = \frac{|S^0_b \cap T^0| + e_b}{N_0}.$$

It follows that $\|p^0\widehat{P}^0 - p^0\|_{\mathrm{TV}} \leq q/N_0$. Replacing only the tail row of $\widehat{P}^0$ by ν changes the product measure by at most p^0_* in total variation. Use (57) to obtain (65).

The minorization gives

$$\|\mu\overline{P}^0 - \mu'\overline{P}^0\|_{\mathrm{TV}} \leq (1-\alpha)\|\mu - \mu'\|_{\mathrm{TV}}, \tag{67}$$

since every row can be written as $\alpha\nu + (1-\alpha)Q_a$. Apply Lemma 7.7 with $P = \overline{P}^0$, $p = p^0$, and $\zeta = C(\rho + q/N_0)$. Since α is bounded away from zero, its factor is absorbed into C, proving (66). □

7.4 Residual-set invariant during terminal exposure

After u compatible terminal-window edges have been exposed, let $S^{(u)}$ and $T^{(u)}$ be the current global residual source and target sets. For each ℓ, let $\xi^{(u)}_\ell$ be the current tip of the ℓth path: initially $\xi^{(0)}_\ell = v_{\ell,r_\ell-m}$; whenever an edge of that terminal window is exposed, replace its current tip by the newly chosen target. A terminal history is called *compatible* if no chosen target is an already exposed root and all exposed assignments remain consistent.

Lemma 7.9 (Evolving residual-set invariant)**.** *After every compatible terminal history,*

$$T^{(u)} \setminus S^{(u)} = \{I_1, \dots, I_q\}, \qquad S^{(u)} \setminus T^{(u)} = \{\xi_1^{(u)}, \dots, \xi_q^{(u)}\}. \tag{68}$$

Consequently,

$$T^{(u)} \cap (S^0 \setminus S^{(u)}) = \varnothing. \tag{69}$$

In particular, every currently available target that lies in S^0 is also a currently available source.

Proof. At $u = 0$, (68) is exactly (52). Suppose it holds before exposing an edge from the current tip $x = \xi_\ell^{(u)}$. Then $x \in S^{(u)} \setminus T^{(u)}$. Compatibility excludes targets in $T^{(u)} \setminus S^{(u)}$, which are precisely the roots by the induction hypothesis. Hence the chosen target y lies in $S^{(u)} \cap T^{(u)}$. Passing to $S^{(u+1)} = S^{(u)} \setminus \{x\}$ and $T^{(u+1)} = T^{(u)} \setminus \{y\}$ replaces the tip x by y and leaves the root defect unchanged. This proves the induction. Finally, any point of the left side of (69) would belong to $T^{(u)} \setminus S^{(u)}$ and to S^0, but the former set is the root set and the roots lie in $T^0 \setminus S^0$. □

7.5 Adaptive comparison during the terminal window

The rows remaining after the prefixes are still independent permanental matchings by Lemma 5.1. During all q terminal windows together, at most qm further source-target pairs are exposed. Let $\mathcal{F}_{\mathrm{pre}}$ be the sigma-field generated by $\mathcal{H}$, the roots, and all exact source-target assignments in the frozen prefixes.

Lemma 7.10 (One terminal endpoint is almost uniform)**.** *Condition on a realization of $\mathcal{F}_{\mathrm{pre}}$ corresponding to a compatible, disjoint prefix history on which $\mathcal{G}_n \cap \mathcal{D}_{n,\mathbf{r}}$ holds. Condition further on any previously exposed compatible terminal history. Consider one of the remaining terminal windows and suppose its starting row is finite. Then its endpoint after the m remaining edges is at total-variation distance at most $\varepsilon_{n,q}$ from U_n, where*

$$\varepsilon_{n,q} \le C_q \left[(1-\alpha)^{m-1} + mh + \frac{m^2}{M} + m\rho + \frac{m \log n}{M} + \rho + \frac{K}{n} \right]. \tag{70}$$

The bound is uniform in the previously exposed compatible history and in the chosen lengths.

Proof. We compare a stopped version of the actual row process to the fixed reset chain $\overline{P}^0$. The stopped process enters a cemetery state † if an intermediate edge chooses a target in $T^{(u)} \setminus S^{(u)}$, equivalently one of the exposed roots. Whenever this happens, complete the stopped transition artificially by an independent draw from ν, but mark the completion as a coupling failure.

Suppose the current source is in a finite row a, and put $T_a^{(u)} := T^{(u)} \cap T_a^0$. At most qm targets have been removed from T_a^0; by (54) and (48), $|T_a^{(u)}| \ge m_a^0 - qm \ge cM$ for all large n. Moreover,

$$\| \operatorname{Unif}(T_a^{(u)}) - \operatorname{Unif}(T_a^0) \|_{\mathrm{TV}} \le \frac{qm}{m_a^0} \le C_q \frac{m}{M}.$$

By hereditary flatness, Lemma 5.2, the actual one-edge law is at distance at most Ch from $\operatorname{Unif}(T_a^{(u)})$. By Lemma 7.9, every target in $T_a^{(u)} \cap S^0$ is still a current source, so pushing such a target to its source-row label is well defined. A uniform target in $T_a^0 \cap S_b^0$ produces state b, whereas a uniform target in $T_a^0 \setminus S^0$ is reset according to ν; this is exactly $\widehat{P}^0(a, \cdot)$. The latter

defect set consists of the roots lying in row a and has cardinality at most q. By the upper bound in (26),

$$\mathbb{P}(\dagger \mid \text{current history}) \le \frac{e^{\kappa h} q}{|T_a^{(u)}|} \le \frac{C_q}{M}.$$

Consequently, by the preceding total-variation comparisons and data processing, the completed actual next-row law can be coupled to $\widehat{P}^0(a,\cdot)$, while charging every cemetery completion as failure, with total one-step error

$$\eta_{n,q} \le C_q \left(h + \frac{m}{M} \right). \tag{71}$$

For a finite row, (56) and the fact that the completion in (61) is supported on early states give

$$\widehat{P}^0(a,*) \le C \left(\rho + \frac{\log n}{M} \right) =: \tau_n. \tag{72}$$

Couple each completed transition maximally to the reset chain while the common state is finite. Declare failure at a mismatch, at a cemetery completion, or upon a move to the tail state. Over the first $m-1$ row transitions, the failure probability is at most

$$(m-1)(\eta_{n,q} + \tau_n). \tag{73}$$

Before the first failure, the completed actual row process and the reset chain agree. Lemma 7.7, with $\beta_{m-1} \le (m-1)(\eta_{n,q} + \tau_n)$, therefore couples the completed actual row after $m-1$ transitions to a reference row $A_0 \sim p^0$ with failure probability at most

$$(m-1)(\eta_{n,q} + \tau_n) + (1-\alpha)^{m-1} + C \left(\rho + \frac{q}{N_0} \right).$$

If $A_0 = a \le d$ and the coupling has not failed, hereditary flatness and the deletion of at most qm frozen targets show that the final target law is within $C_q(h + m/M)$ of λ_a^0. A final hit on an exposed root is marked as failure and is absorbed in the same bound. If $A_0 = *$, draw an auxiliary target from λ_*^0 and mark the tail-row event as failure. The reference target law is then exactly

$$\sum_a p_a^0 \lambda_a^0 = \mathrm{Unif}(T^0). \tag{74}$$

The tail-row completion costs $p_*^0 = O(\rho)$ by (57). Finally, $|[n] \setminus T^0| \le K$, so

$$\| \mathrm{Unif}(T^0) - U_n \|_{\mathrm{TV}} \le K/n. \tag{75}$$

Combining (71)–(75), using $q/N_0 \le C_q K/n$, gives (70). Its convergence to zero follows from Definition 7.1 and $K/n = \delta \to 0$. □

Corollary 7.11 (Linear-depth polynomial finite-endpoint mixing)**.** *Let $L = \log n$. Fix $q \ge 1$ and $Q > 0$. There is $A = A(\kappa, q, Q, \tau_0) < \infty$ such that, whenever*

$$L^3 \le r_- \le r_+ \le \frac{n}{L^A},$$

the choice

$$m = \lfloor L_\delta \rfloor \tag{76}$$

is admissible and

$$\varepsilon_{n,q} \le C_{\kappa,q,Q} L^{-Q}, \tag{77}$$

$$\Delta_{n,q} \le C_{\kappa,q,Q} L^{-Q}. \tag{78}$$

The conditional product bound in Corollary 7.12 is at most $C_{\kappa,q,Q} L^{-Q}$ as well.

Proof. Since $r_- \ge L^3$ and $L_\delta \le L + O_q(1)$, one has $m \le r_-/3$ for all large n. Also

$$(1-\alpha)^{m-1} \le C\delta^c, \qquad mh \le C(1+L_\delta)\delta^{1/4}, \qquad m\rho \le C(1+L_\delta)\delta^{1/(4\kappa)}$$

for a constant $c = c(\kappa, \tau_0) > 0$. Furthermore, $M = \sqrt{nK} \ge c_q \sqrt{n}\, L^{3/2}$ and $m \le L + O_q(1)$, so

$$\frac{m^2 + m\log n}{M} \le C_q \frac{L^{1/2}}{\sqrt{n}}, \qquad \frac{m}{hM} \le C_q \frac{1}{n^{1/4} L^{5/4}}.$$

Thus m is admissible. Since $\delta = qr_+/n \le qL^{-A}$, equation (70) gives (77) after choosing A sufficiently large.

For the global estimate, Corollary 4.3 gives $\zeta_n \le C_\kappa r_+/n$. The depletion term is $O_q(\delta^{1/2} L_\delta)$ by Lemma 7.5. The environment error in (34) is smaller than every fixed power of L^{-1} uniformly in the stated range, because $M \ge c_q \sqrt{n}\, L^{3/2}$ and $hM = n\delta^{3/4} \ge c_q n^{1/4} L^{9/4}$. Substituting these estimates into (46) proves (78) after increasing A once more. □

Corollary 7.12 (Several terminal endpoints)**.** *On the $\mathcal{F}_{\mathrm{pre}}$-measurable event that the prefix history is compatible and disjoint, $\mathcal{G}_n \cap \mathcal{D}_{n,\mathbf{r}}$ holds, and all terminal-window starting rows are finite, the regular conditional law satisfies*

$$\left\| \mathcal{L}\left((\sigma_n^{r_\ell}(I_\ell))_{\ell=1}^q \;\middle|\; \mathcal{F}_{\mathrm{pre}} \right) - \mathrm{Unif}([n])^{\otimes q} \right\|_{\mathrm{TV}} \le q\varepsilon_{n,q}. \tag{79}$$

Equivalently, the endpoint vector can be coupled to a product-uniform vector $(J_1, \dots, J_q)$ that is independent of $\mathcal{F}_{\mathrm{pre}}$, with failure probability at most $q\varepsilon_{n,q}$.

Proof. Expose the terminal windows successively and apply Lemma 7.10 conditionally at each stage. Its estimates are uniform in every previously exposed compatible terminal history. Across all windows at most qm additional assignments are exposed; by (48), these deletions cannot destroy the row-size or early-cell margins obtained after the prefixes. On the success event, Lemma 7.9 preserves the residual-set bookkeeping needed for the next window. Every hit on an exposed root, including on a final edge, is already charged as a coupling failure in Lemma 7.10. At stage ℓ, use a fresh uniform coordinate independent of $\mathcal{F}_{\mathrm{pre}}$ and of the preceding coordinates. The sequential coupling and a union bound prove (79). □

7.6 Averaging down to the coarse environment

Proof of Proposition 7.2. Fix $\mathbf{r} \in [r_-, r_+]^q$ and abbreviate $E_\ell = E_\ell(\mathbf{r})$. Let $\Gamma_{n,\mathbf{r}}$ be the $\mathcal{F}_{\mathrm{pre}}$-measurable event that the frozen prefixes are open and pairwise disjoint, $\mathcal{G}_n \cap \mathcal{D}_{n,\mathbf{r}}$ holds, and every terminal-window starting row is finite. Put

$$d_{n,q} := \sup_{\mathbf{r} \in [r_-, r_+]^q} \mathbb{P}(\mathcal{D}_{n,\mathbf{r}}^c).$$

By Lemma 7.5,

$$d_{n,q} \le C_q \left[\delta^{3/4} L_\delta + \delta^{1/2} L_\delta\right].$$

The estimates above and a union bound give

$$\sup_{\mathbf{r}} \mathbb{P}(\Gamma_{n,\mathbf{r}}^c) \le q\zeta_n + Cq^2 \frac{r_+}{n} + \mathbb{P}(\mathcal{G}_n^c) + d_{n,q} + q\rho. \tag{80}$$

For the last term, conditional on the full clocks and permutation each v_{ℓ,k_ℓ} is uniform on $[n]$, while $\mathbb{E}(m_*/n) \le \rho$.

On $\Gamma_{n,\mathbf{r}}$, Corollary 7.12 gives

$$\left\|\mathcal{L}(E_1, \dots, E_q \mid \mathcal{F}_{\mathrm{pre}}) - U_n^{\otimes q}\right\|_{\mathrm{TV}} \le q\varepsilon_{n,q};$$

on its complement the distance is at most one. Since $\sigma(\mathcal{H}, I_1, \dots, I_q) \subset \mathcal{F}_{\mathrm{pre}}$, convexity of total variation under conditional averaging yields

$$\begin{aligned}&\left\|\mathcal{L}(E_1, \dots, E_q \mid \mathcal{H}, I_1, \dots, I_q) - U_n^{\otimes q}\right\|_{\mathrm{TV}} \\ &\quad \le \mathbb{E}\left[\left\|\mathcal{L}(E_1, \dots, E_q \mid \mathcal{F}_{\mathrm{pre}}) - U_n^{\otimes q}\right\|_{\mathrm{TV}} \mid \mathcal{H}, I_1, \dots, I_q\right].\end{aligned}$$

Taking expectations, then the supremum over $\mathbf{r}$, gives

$$\Delta_{n,q} \le q\varepsilon_{n,q} + q\zeta_n + Cq^2 \frac{r_+}{n} + \mathbb{P}(\mathcal{G}_n^c) + d_{n,q} + q\rho. \tag{81}$$

Use (28) and $K/n = \delta$ to absorb the short-root and intersection terms into $C_q\delta$. Then substitute (70), (34), and (50); since $\delta^{3/4} L_\delta \le \delta^{1/2} L_\delta$ for small δ, this is exactly (46). Every term tends to zero by Definition 7.1, proving (45). □

Corollary 7.13 (Bounded coarse observables)**.** *Let*

$$G_n = G_n(\mathcal{H}; i_1, \dots, i_q, j_1, \dots, j_q)$$

be a real- or complex-valued measurable function with $\|G_n\|_\infty \le B < \infty$ uniformly in n. Let $J_1, \dots, J_q$ be independent U_n labels, independent of the clocks and roots. Then

$$\begin{aligned}\sup_{\mathbf{r} \in [r_-, r_+]^q} &\big|\mathbb{E} G_n(\mathcal{H}; I_1, \dots, I_q, E_1(\mathbf{r}), \dots, E_q(\mathbf{r})) \\ &- \mathbb{E} G_n(\mathcal{H}; I_1, \dots, I_q, J_1, \dots, J_q)\big| \le 2B\Delta_{n,q} \longrightarrow 0.\end{aligned} \tag{82}$$

Proof. Condition on $\mathcal{H}, I_1, \dots, I_q$ and use the dual bound $|\int g \, \mathrm{d}\mu - \int g \, \mathrm{d}\nu| \le 2\|g\|_\infty \|\mu - \nu\|_{\mathrm{TV}}$, then apply Proposition 7.2. □

7.7 Recovering observables of the exact clocks

For each finite block choose a deterministic representative $\bar{t}_a \in B_a$ and put $\bar{X}_i = \bar{t}_a$ for $i \in H_a$. When i, j are in finite blocks, define

$$\overline{Z}_{ij} := -(\theta_i - \theta_j)(\bar{X}_j - \bar{X}_i). \tag{83}$$

Then

$$|\log R_{ij} - \overline{Z}_{ij}| \le 2\kappa h \tag{84}$$

whenever both labels are finite.

Corollary 7.14 (Bounded Lipschitz exact-clock observables)**.** *For $1 \le \ell \le q$, let $\psi_{n,\ell} : [n]^2 \times \mathbb{R} \to \mathbb{C}$ satisfy, uniformly in n, ℓ, i, j, z, z',*

$$|\psi_{n,\ell}(i,j,z)| \le B, \qquad |\psi_{n,\ell}(i,j,z) - \psi_{n,\ell}(i,j,z')| \le L|z - z'|. \tag{85}$$

Define

$$\Psi_{n,\ell}(i,j) := \psi_{n,\ell}(i,j,\log R_{ij}).$$

Then, uniformly in $\mathbf{r} \in [r_-, r_+]^q$,

$$\left| \mathbb{E} \prod_{\ell=1}^{q} \Psi_{n,\ell}(I_\ell, E_\ell(\mathbf{r})) - \mathbb{E} \prod_{\ell=1}^{q} \Psi_{n,\ell}(I_\ell, J_\ell) \right| \longrightarrow 0, \tag{86}$$

$$\left| \mathbb{E} \prod_{\ell=1}^{q} \left[\mathbf{1}_{\{L_n(I_\ell) > r_\ell\}} \Psi_{n,\ell}(I_\ell, E_\ell(\mathbf{r})) \right] - \mathbb{E} \prod_{\ell=1}^{q} \Psi_{n,\ell}(I_\ell, J_\ell) \right| \longrightarrow 0. \tag{87}$$

The functions may depend on the deterministic rates and on arbitrary bounded deterministic marks through their (i,j) arguments.

Proof. Set

$$\overline{\Psi}_{n,\ell}(i,j) := \begin{cases} \psi_{n,\ell}(i,j,\overline{Z}_{ij}), & i,j \notin H_*, \\ 0, & \text{otherwise.} \end{cases}$$

The product of the $\overline{\Psi}_{n,\ell}$ is a bounded coarse observable. Corollary 7.13 therefore replaces the actual endpoint vector by $(J_1, \dots, J_q)$. On the event that all labels are finite, (84), (85), and a telescoping product estimate show that the coarse-to-exact replacement costs at most $C_{q,B,L,\kappa} h$. Each root, actual endpoint, and fresh endpoint is marginally uniform on $[n]$ conditional on the full permutation where appropriate, so the probability that any required label lies in H_* is $O_q(\rho)$. This proves (86). Finally, the probability that one of the long-cycle indicators vanishes is at most $q\zeta_n$, proving (87). □

Corollary 7.15 (Truncated swap-factor decorrelation)**.** *Fix $1 \le A_1, \dots, A_q < \infty$. Uniformly in $\mathbf{r} \in [r_-, r_+]^q$,*

$$\mathbb{E} \prod_{\ell=1}^{q} \left[\mathbf{1}_{\{L_n(I_\ell) > r_\ell\}} F_{A_\ell}(I_\ell, E_\ell(\mathbf{r})) \right] - \mathbb{E} \prod_{\ell=1}^{q} F_{A_\ell}(I_\ell, J_\ell) \longrightarrow 0. \tag{88}$$

Proof. Apply Corollary 7.14 with $\psi_{n,\ell}(i,j,z) = e^z \wedge A_\ell$, which is bounded by A_ℓ and A_ℓ-Lipschitz. □

Corollary 7.16 (Quantitative truncation-uniform observable interface)**.** *In the setting of Proposition 7.2, fix $q \in \{1,2\}$ and use an admissible free terminal depth. Let $\mathcal{G}_{n,q}$ denote the bracket on the right side of* (46)*. For every truncation level $A \ge 1$, which may depend on n, and every $r \in [r_-, r_+]$, put*

$$S_{n,r}^{(A)} = \sum_{i : L_n(i) > r} (R_{i,\sigma_n^r(i)} \wedge A), \qquad D_{n,A} = \frac{1}{n^2} \sum_{i,j} (R_{ij} \wedge A).$$

The one-root construction gives

$$\left|\mathbb{E}[S_{n,r}^{(A)}/n] - \mathbb{E}D_{n,A}\right| \le C_\kappa A \mathcal{G}_{n,1}, \tag{89}$$

and the two-root construction gives

$$\left|\mathbb{E}[(S_{n,r}^{(A)}/n)^2] - \mathbb{E}D_{n,A}^2\right| \le C_\kappa A^2 \mathcal{G}_{n,2}. \tag{90}$$

The constants are independent of A. These are annealed observable estimates, not an arbitrary-deletion quenched comparison.

Proof. The function $z \mapsto e^z \wedge A$ is bounded by A and is A-Lipschitz. On finite blocks, (84) therefore bounds a one-factor coarsening error by $2\kappa A h$. A telescoping product bounds the two-factor error by $C_\kappa A^2 h$. Apply the quantitative comparison of Proposition 7.2 to the bounded coarse test, and then undo this coarsening. The tail-label exceptions have probability $O_q(\rho)$. The internal root bound in Corollary 4.3 gives $\mathbb{P}\{L_n(I_\ell) \le r_+\} = O_\kappa(r_+/n)$. Multiplying these probabilities by A^q charges both kinds of exception to the displayed bracket, since $h, \rho, r_+/n$ are already absorbed there. All estimates remain linear in the one-factor bound and quadratic in the two-factor bound, so no fixed-A limiting argument is used.

Conditional on the clocks, independent uniform roots turn the selected one-factor expectation into $S_{n,r}^{(A)}/n$ and the two-factor expectation into $(S_{n,r}^{(A)}/n)^2$. Independent fresh uniform pairs similarly give $D_{n,A}$ and $D_{n,A}^2$. The fresh pairs share the clock environment: the latter quantity is $\mathbb{E}D_{n,A}^2$, not $(\mathbb{E}D_{n,A})^2$. This proves both claims. □

Corollary 7.17 (Untruncated swap-factor decorrelation below the moment threshold)**.** *If the fixed integer q satisfies $q < \lambda_\kappa$, then uniformly in $\mathbf{r} \in [r_-, r_+]^q$,*

$$\left|\mathbb{E}\prod_{\ell=1}^{q}\left[\mathbf{1}_{\{L_n(I_\ell)>r_\ell\}} R_{I_\ell,E_\ell(\mathbf{r})}\right] - \mathbb{E}\prod_{\ell=1}^{q} R_{I_\ell,J_\ell}\right| \longrightarrow 0. \tag{91}$$

No factorization of the fresh-pair expectation is asserted.

Proof. Choose p with $q < p < \lambda_\kappa$. By (16) and (17), every actual selected factor and every fresh-pair factor is uniformly bounded in L^p. Generalized Hölder therefore makes each of the two products uniformly integrable: for example, the contribution of $\max_\ell Z_\ell > A$ is $O(A^{-(p-q)})$ after increasing the constant, where Z_ℓ denotes the corresponding nonnegative factor. Apply Corollary 7.15 with all truncation levels equal to A, then let first $n \to \infty$ and afterward $A \to \infty$. □

8 Concentration of the switching intensity

Define the truncated intensity

$$S_{n,r}^{(A)} := \sum_{i:L_n(i)>r} F_A(i, \sigma_n^r(i)). \tag{92}$$

Also put

$$D_{n,A} := \frac{1}{n^2}\sum_{i,j=1}^{n} F_A(i,j). \tag{93}$$

Conditional on the clocks, $D_{n,A} = \mathbb{E}(F_A(I,J) \mid X)$.

For completeness we record the elementary variance inequality used below. If $Z = g(Y_1, \dots, Y_n)$ for independent inputs and $Z^{(k)}$ is obtained by replacing Y_k by an independent copy, then

$$\operatorname{Var} Z \le \sum_{k=1}^{n} \mathbb{E}(Z - Z^{(k)})^2. \tag{94}$$

Indeed, for $\mathcal{F}_k = \sigma(Y_1, \dots, Y_k)$ and $D_k = \mathbb{E}(Z \mid \mathcal{F}_k) - \mathbb{E}(Z \mid \mathcal{F}_{k-1})$, martingale orthogonality gives $\operatorname{Var} Z = \sum_k \mathbb{E} D_k^2$. Also $D_k = \mathbb{E}(Z - Z^{(k)} \mid \mathcal{F}_k)$, so conditional Jensen proves (94). The sharper factor-1/2 form is the standard Efron–Stein inequality; see Boucheron–Lugosi–Massart [1, Theorem 3.1].

Lemma 8.1 (The independent-pair average concentrates)**.** *For every fixed* $1 \le A < \infty$*,*

$$\operatorname{Var}(D_{n,A}) \le \frac{4A^2}{n}. \tag{95}$$

Proof. Replacing one clock X_k changes only summands in (93) with $i = k$ or $j = k$, at most $2n$ terms. Each term lies in $[0, A]$, so $|D_{n,A} - D_{n,A}^{(k)}| \le 2A/n$. Applying (94) gives

$$\operatorname{Var}(D_{n,A}) \le n \left(\frac{2A}{n} \right)^2 = \frac{4A^2}{n}.$$

□

Proposition 8.2 (Truncated switching intensity)**.** *For every fixed* $1 \le A < \infty$*,*

$$\sup_{r_- \le r \le r_+} \mathbb{E} \left| \frac{S_{n,r}^{(A)}}{n} - \mu_{n,A} \right|^2 \longrightarrow 0. \tag{96}$$

Proof. Let I be uniform. Conditional on the clocks,

$$\frac{S_{n,r}^{(A)}}{n} = \mathbb{E} \left[\mathbf{1}_{\{L_n(I) > r\}} F_A(I, \sigma_n^r(I)) \;\middle|\; X \right].$$

The $q = 1$ case of Corollary 7.15 gives

$$\mathbb{E} \frac{S_{n,r}^{(A)}}{n} - \mathbb{E} D_{n,A} \longrightarrow 0$$

uniformly in r. For independent uniform roots I_1, I_2 and independent uniform labels J_1, J_2, all independent of the clocks,

$$\mathbb{E} \left(\frac{S_{n,r}^{(A)}}{n} \right)^2 = \mathbb{E} \prod_{\ell=1}^{2} \left[\mathbf{1}_{\{L_n(I_\ell) > r\}} F_A\big(I_\ell, \sigma_n^r(I_\ell)\big) \right],$$

$$\mathbb{E} D_{n,A}^2 = \mathbb{E} \prod_{\ell=1}^{2} F_A(I_\ell, J_\ell).$$

Thus the $q = 2$ case of the same corollary gives

$$\mathbb{E}\left(\frac{S_{n,r}^{(A)}}{n}\right)^2 - \mathbb{E}D_{n,A}^2 \longrightarrow 0$$

uniformly in r. Since $\mathbb{E}D_{n,A} = \mu_{n,A}$ and $\mathrm{Var}(D_{n,A}) \to 0$ by Lemma 8.1, the first two moments of $S_{n,r}^{(A)}/n$ converge uniformly to $\mu_{n,A}$ and $\mu_{n,A}^2$, respectively. This is (96). □

Corollary 8.3 (Mixed truncated switching moments)**.** *Fix $q \geq 1$ and $1 \leq A_1, \ldots, A_q < \infty$. Then*

$$\sup_{\mathbf{r}\in[r_-,r_+]^q} \left| \mathbb{E}\prod_{\ell=1}^{q} \frac{S_{n,r_\ell}^{(A_\ell)}}{n} - \prod_{\ell=1}^{q} \mu_{n,A_\ell} \right| \longrightarrow 0. \tag{97}$$

Proof. Conditional on the clocks, independent uniform roots give

$$\mathbb{E}\prod_{\ell=1}^{q} \frac{S_{n,r_\ell}^{(A_\ell)}}{n} = \mathbb{E}\prod_{\ell=1}^{q} \left[\mathbf{1}_{\{L_n(I_\ell)>r_\ell\}} F_{A_\ell}(I_\ell, \sigma_n^{r_\ell}(I_\ell))\right].$$

Corollary 7.15 replaces the right side by $\mathbb{E}\prod_{\ell=1}^{q} D_{n,A_\ell} + o(1)$, uniformly in $\mathbf{r}$. Each D_{n,A_ℓ} is bounded by A_ℓ, and Lemma 8.1 gives $\mathbb{E}|D_{n,A_\ell} - \mu_{n,A_\ell}| = O(n^{-1/2})$. A telescoping product estimate proves (97). □

Theorem 8.4 (Annealed switching intensity)**.** *For every mesoscopic window $[r_-, r_+]$,*

$$\sup_{r_-\leq r\leq r_+} \mathbb{E}\left|\frac{S_{n,r}}{n} - 1\right| \longrightarrow 0. \tag{98}$$

Proof. For $A \geq 1$,

$$\begin{aligned}\mathbb{E}\left|\frac{S_{n,r}}{n} - 1\right| &\leq \mathbb{E}\left|\frac{S_{n,r}^{(A)}}{n} - \mu_{n,A}\right| + |1 - \mu_{n,A}| \\ &\quad + \frac{1}{n}\mathbb{E}(S_{n,r} - S_{n,r}^{(A)}).\end{aligned}$$

For fixed A, the first term tends to zero uniformly in r by Proposition 8.2. The second term is bounded by (24), and the third by Lemma 4.1. First let $n \to \infty$, then $A \to \infty$. □

Corollary 8.5 (Higher moments of the switching intensity)**.** *For every $1 \leq p < \lambda_\kappa$,*

$$\sup_{r_-\leq r\leq r_+} \mathbb{E}\left|\frac{S_{n,r}}{n} - 1\right|^p \longrightarrow 0. \tag{99}$$

In particular, the convergence is uniform in L^2 whenever $\kappa < 2$.

Proof. The case $p = 1$ is Theorem 8.4. Fix $1 < p < \lambda_\kappa$ and choose u with $p < u < \lambda_\kappa$. If I is uniform and independent of the clocks, conditional Jensen and Corollary 4.2 give, uniformly in n, r,

$$\begin{aligned}\mathbb{E}\left(\frac{S_{n,r}}{n}\right)^u &= \mathbb{E}\left(\mathbb{E}\left[\mathbf{1}_{\{L_n(I)>r\}} R_{I,\sigma_n^r(I)} \;\middle|\; X\right]\right)^u \\ &\leq \frac{1}{n}\mathbb{E}\sum_{i:L_n(i)>r} R_{i,\sigma_n^r(i)}^u \leq C_{\kappa,u}.\end{aligned}$$

Thus $S_{n,r}/n - 1$ is uniformly bounded in L^u. Interpolation between L^1 and L^u, together with Theorem 8.4, proves (99). □

Corollary 8.6 (Mixed untruncated switching moments below the threshold)**.** *If the fixed integer q satisfies $q < \lambda_\kappa$, then*

$$\sup_{\mathbf{r}\in[r_-,r_+]^q} \left| \mathbb{E} \prod_{\ell=1}^{q} \frac{S_{n,r_\ell}}{n} - 1 \right| \longrightarrow 0. \tag{100}$$

Proof. Put $Y_{n,r} = S_{n,r}/n$. Corollary 8.5 gives $\sup_r \|Y_{n,r} - 1\|_q \to 0$ and $\sup_{n,r} \|Y_{n,r}\|_q < \infty$. A telescoping product identity and Hölder's inequality yield, uniformly in $\mathbf{r}$,

$$\mathbb{E} \left| \prod_{\ell=1}^{q} Y_{n,r_\ell} - 1 \right| \le \sum_{\ell=1}^{q} \|Y_{n,r_\ell} - 1\|_q \prod_{k<\ell} \|Y_{n,r_k}\|_q,$$

which tends to zero. □

Proposition 8.7 (Bounded weighted switching intensity)**.** *Let $a^{(n)} = (a_{n,ij})_{i,j\in[n]}$ be deterministic real- or complex-valued arrays with $\sup_n \max_{i,j} |a_{n,ij}| < \infty$. Then*

$$\sup_{r_-\le r\le r_+} \left| \frac{1}{n} \mathbb{E} \sum_{i:L_n(i)>r} R_{i,\sigma_n^r(i)} a_{n,i,\sigma_n^r(i)} - \frac{1}{n^2} \sum_{i,j=1}^{n} a_{n,ij} \right| \longrightarrow 0. \tag{101}$$

Proof. Let $B = \sup_n \max_{i,j} |a_{n,ij}|$. For fixed $A \ge 1$, apply Corollary 7.14 with $q = 1$ and

$$\psi_{n,1}(i,j,z) = a_{n,ij}(e^z \wedge A).$$

This gives, uniformly in r,

$$\begin{aligned} \frac{1}{n} \mathbb{E} \sum_{i:L_n(i)>r} & a_{n,i,\sigma_n^r(i)} F_A(i, \sigma_n^r(i)) \\ & - \mathbb{E}\big[a_{n,I,J} F_A(I,J)\big] \longrightarrow 0. \end{aligned}$$

For $\kappa > 1$, the discarded actual tail is at most $BC_\kappa A^{-1/(\kappa-1)}$ by Lemma 4.1, while the discarded fresh-pair tail satisfies the same bound by (22). For $\kappa = 1$ both tails vanish. Finally,

$$\mathbb{E}[a_{n,I,J} R_{I,J}] = \frac{1}{n^2} \sum_{i,j=1}^{n} a_{n,ij}$$

by (21). Let first $n \to \infty$ and then $A \to \infty$. □

Corollary 8.8 (Marked mesoscopic-cycle first moments)**.** *Let $b^{(n)} = (b_{n,j})_{j\in[n]}$ be deterministic real- or complex-valued marks with $\sup_n \max_j |b_{n,j}| < \infty$, and define*

$$C_{n,r}[b] := \sum_{\substack{C\in \mathrm{Cyc}(\sigma_n) \\ |C|=r}} \frac{1}{r} \sum_{j\in C} b_{n,j}. \tag{102}$$

Then

$$\sup_{r_-\le r\le r_+} \left| r\mathbb{E} C_{n,r}[b] - \frac{1}{n} \sum_{j=1}^{n} b_{n,j} \right| \longrightarrow 0. \tag{103}$$

Proof. Apply the marked identity (9), by linearity, with $\Psi(i,j,\sigma) = b_{n,j}$. Since every r-cycle has $n-r$ labels outside it,

$$r(n-r)\mathbb{E}C_{n,r}[b] = \mathbb{E}\sum_{i:L_n(i)>r} R_{i,\sigma_n^r(i)} b_{n,\sigma_n^r(i)}.$$

Use Proposition 8.7 with $a_{n,ij} = b_{n,j}$ and $r_+ = o(n)$. □

Proof of Theorem 1.1(i). By the exact identity (12) and Theorem 8.4, uniformly for $r \in [r_-, r_+]$,

$$r\mathbb{E}C_{n,r} = \frac{n}{n-r}\frac{\mathbb{E}S_{n,r}}{n} = 1 + o(1),$$

because $r_+ = o(n)$. □

9 Exponential tilting and the Poisson point process

Fix $a_n \to \infty$ with $a_n = o(n)$, and let $f \in C_c^+((0,\infty))$. Choose $0 < u < v < \infty$ with $\operatorname{supp} f \subset [u,v]$. Set

$$f_{n,r} := f(r/a_n), \qquad Y_n := \sum_{r\geq 1} f_{n,r} C_{n,r} = \langle f, \Xi_n \rangle, \tag{104}$$

and

$$\mathcal{L}_n(t) := \mathbb{E}e^{-tY_n}, \qquad \frac{\mathrm{d}\mathbb{P}_{n,t}}{\mathrm{d}\mathbb{P}} := \frac{e^{-tY_n}}{\mathcal{L}_n(t)}, \qquad 0 \leq t \leq 1. \tag{105}$$

Write $\mathbb{E}_t$ for expectation under $\mathbb{P}_{n,t}$.

Lemma 9.1 (Uniform lower bound for the tilted partition function)**.** *There is $c_f > 0$ such that*

$$\inf_n \inf_{0\leq t\leq 1} \mathcal{L}_n(t) \geq c_f. \tag{106}$$

Proof. By Theorem 1.1(i), uniformly for $ua_n \leq r \leq va_n$, $\mathbb{E}C_{n,r} \leq C/r$ for all large n. Hence

$$\mathbb{E}Y_n \leq \|f\|_\infty \sum_{ua_n\leq r\leq va_n} \frac{C}{r} \leq C_f.$$

Jensen's inequality gives $\mathcal{L}_n(t) = \mathbb{E}e^{-tY_n} \geq e^{-t\mathbb{E}Y_n} \geq e^{-C_f}$. Absorb finitely many small n into the constant. □

Proposition 9.2 (Tilted one-cycle intensity)**.** *Uniformly for $0 \leq t \leq 1$ and for integers $ua_n \leq r \leq va_n$,*

$$r\mathbb{E}_t C_{n,r} = e^{-tf_{n,r}} + o(1). \tag{107}$$

Proof. Apply Proposition 3.2 with $\Phi(\sigma) = e^{-tY_n(\sigma)}$. If $j = \sigma^r(i)$ and $L = L_n(i) > r$, then right-composition by $(i\,j)$ splits the L-cycle into cycles of lengths r and $L-r$. Therefore

$$Y_n(\sigma \circ (i\,j)) - Y_n(\sigma) = f_{n,r} + f_{n,L-r} - f_{n,L}. \tag{108}$$

If $L > r + va_n$, then $L > va_n$ and $L - r > va_n$, so the last two terms in (108) vanish. On this event,

$$e^{-tY_n(\sigma\circ(i\,j))} = e^{-tf_{n,r}} e^{-tY_n(\sigma)}.$$

Thus

$$\begin{aligned}\Big| r(n-r)\mathbb{E}[C_{n,r}e^{-tY_n}] - e^{-tf_{n,r}}\mathbb{E}[e^{-tY_n}S_{n,r}]\Big| \\ \le 2\mathbb{E}\sum_{i:\,r<L_n(i)\le r+va_n} R_{i,\sigma_n^r(i)}.\end{aligned} \tag{109}$$

The factor 2 is harmless and comes from bounding both exponential weights by one.

The right side is $o(n)$ uniformly in r. Indeed, for fixed A it is at most

$$2An\mathbb{P}(L_n(I_n) \le 2va_n) + 2\mathbb{E}\sum_{i:L_n(i)>r} R_{i,\sigma_n^r(i)}\mathbf{1}_{\{R_{i,\sigma_n^r(i)}>A\}}.$$

The first term is $o(n)$ by Corollary 4.3, since $a_n = o(n)$. When $\kappa > 1$, the second term divided by n is bounded by $C_\kappa A^{-1/(\kappa-1)}$ by Lemma 4.1; let first $n \to \infty$, then $A \to \infty$. When $\kappa = 1$, all swap factors equal one and the second term vanishes for every $A \ge 1$.

Furthermore, Theorem 8.4 gives, uniformly in t, r,

$$\left|\frac{1}{n}\mathbb{E}[e^{-tY_n}S_{n,r}] - \mathcal{L}_n(t)\right| \le \mathbb{E}\left|\frac{S_{n,r}}{n} - 1\right| = o(1). \tag{110}$$

Divide (109) by $n\mathcal{L}_n(t)$, use Lemma 9.1, and note that $(n-r)/n = 1 + o(1)$ uniformly on the window. This proves (107). □

Proof of Theorem 1.1(ii). Since Y_n is a finite sum, $\mathcal{L}_n$ is differentiable and

$$-\frac{\mathrm{d}}{\mathrm{d}t}\log\mathcal{L}_n(t) = \mathbb{E}_t Y_n = \sum_{r\ge1} f_{n,r}\mathbb{E}_t C_{n,r}.$$

By Proposition 9.2, uniformly in $t \in [0,1]$,

$$\mathbb{E}_t Y_n = \sum_r \frac{f_{n,r}e^{-tf_{n,r}}}{r} + o(1). \tag{111}$$

Indeed the uniform error in $r\mathbb{E}_t C_{n,r}$ is multiplied by $\sum_{ua_n\le r\le va_n}|f_{n,r}|/r = O_f(1)$. Integrating (111) from 0 to 1 yields

$$\log\mathcal{L}_n(1) = -\sum_r \frac{1}{r}\int_0^1 f_{n,r}e^{-tf_{n,r}}\,\mathrm{d}t + o(1) \tag{112}$$

$$= -\sum_r \frac{1-e^{-f(r/a_n)}}{r} + o(1). \tag{113}$$

The summand is supported on $[ua_n, va_n]$, and the Riemann-sum identity

$$\sum_r \frac{1-e^{-f(r/a_n)}}{r} = \frac{1}{a_n}\sum_r \frac{1-e^{-f(r/a_n)}}{r/a_n} \longrightarrow \int_0^\infty (1-e^{-f(x)})\frac{\mathrm{d}x}{x}$$

gives

$$\mathbb{E}e^{-\langle f,\Xi_n\rangle} \longrightarrow \exp\left\{-\int_0^\infty (1-e^{-f(x)})\frac{\mathrm{d}x}{x}\right\}. \tag{114}$$

For a Poisson random measure with locally finite intensity μ, the Laplace functional is $\exp\{-\int(1-e^{-f})\,\mathrm{d}\mu\}$; hence the right side is the Laplace functional of the Poisson point process with intensity $\mathrm{d}x/x$. On the locally compact Polish space $(0,\infty)$, convergence of Laplace functionals for all $f \in C_c^+((0,\infty))$ is equivalent to vague convergence in distribution of the associated locally finite random measures; see Daley–Vere-Jones [2, Proposition 11.1.VIII]. Therefore

$$\Xi_n \Rightarrow \mathrm{PPP}(\,\mathrm{d}x/x).$$

□

10 Dependency audit, scope, and extensions

The proof of Theorem 1.1 is self-contained under the comparability hypothesis (1). The short-root estimate used in the endpoint and tilting arguments is Corollary 4.3, which follows from the exact master switch and the selected-factor moment bound. In particular, no macroscopic cycle-limit theorem is imported. Every probabilistic and combinatorial estimate specific to the Luce model has been proved in this document. We cite Boucheron–Lugosi–Massart only for the sharper standard form of the variance inequality (the weaker form actually used was proved here), and Daley–Vere-Jones for the general Laplace-functional criterion for vague convergence of random measures.

The endpoint theorem is a fixed-q estimate in mean conditional total variation relative to the coarse sigma-field $\mathcal{H}$. It is not a statement conditional on all exact clocks. The free-depth audit is as follows. The environment and prefix-depletion estimates do not depend on m. The restriction $m \le r_-/3$ is used only to define nonnegative frozen-prefix lengths. The condition $m/(hM) \to 0$ preserves residual row and early-cell margins under at most qm terminal deletions. The requirement $m \to \infty$ controls the reset term, while mh, $m\rho$, m^2/M, and $m\log n/M$ are exactly the accumulated adaptive-coupling errors in (70). No other step uses the former square-root-depth choice. The proof does not give uniformity for r of order n, a residual local law after arbitrary deletions, or unrestricted growth $q = q_n \to \infty$. The fresh-pair observables in Corollary 7.15 share the same random clock environment and should not be factorized before the separate concentration step in Section 8. Finally, untruncated products of two or more swap factors are not claimed for arbitrary κ; Corollaries 7.17 and 8.6 require the natural moment condition $q < \lambda_\kappa$.

Remark 10.1 (Beyond exponential clocks). The conditional matching and endpoint mechanism is not intrinsically exponential. For independent clocks with positive densities $f_{n,i}$, conditioning on the ordered unlabeled values in a block gives assignment weights proportional to

$$\prod_i f_{n,i}(s_{\pi(i)}).$$

The hereditary-flatness argument would follow from a blockwise cross-ratio condition such as

$$\max_{a\le d}\sup_{i,u}\sup_{x,y\in B_a}\left|\log\frac{f_{n,i}(x)f_{n,u}(y)}{f_{n,i}(y)f_{n,u}(x)}\right| \longrightarrow 0.$$

The associated clock-swap Radon–Nikodym factor is

$$R_{ij} = \frac{f_{n,i}(X_j) f_{n,j}(X_i)}{f_{n,i}(X_i) f_{n,j}(X_j)}.$$

A full extension would also require analogues of the core-tail occupancy, localized root-cycle escape, common-support, and selected-factor uniform integrability hypotheses. We record this only as a roadmap, not as a theorem of the present paper.

Remark 10.2 (Further endpoint questions). Two natural strengthenings remain open within the present bookkeeping. For $q = q_n \to \infty$, all constants and bad-event probabilities must be tracked simultaneously; in particular one encounters terms of the forms

$$\frac{q_n^2 r_+}{n}, \qquad q_n m h, \qquad \frac{q_n^2 m^2}{M}, \qquad \frac{q_n m}{hM}, \qquad q_n m \rho,$$

as well as $q_n(1-\alpha)^m$. No growing-q conclusion is asserted here. Likewise, joint observations $\sigma_n^{r_1}(I), \dots, \sigma_n^{r_q}(I)$ along one orbit require a new connected-path exposure argument and are not a corollary of the independent-root theorem. The abstract Doeblin comparison, Proposition 7.2, and the marked switching identity provide reusable inputs for these questions.

References

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